

Def. Given a space  $X$ , a covering space of  $X$  consists of a space  $\tilde{X}$  and a map  $p: \tilde{X} \rightarrow X$  s.t.  
 For each  $x \in X$ , there is an open neighborhood  $U$  of  $x$  in  $X$  satisfying

$$p^{-1}[U] = \bigsqcup_{i \in \mathbb{N}} \mathcal{O}_i \quad \text{and} \quad p|_{\mathcal{O}_i}: \mathcal{O}_i \xrightarrow{\cong} U_i.$$

(Such  $U$  is called "evenly covered")

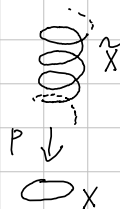
Example. Given a space  $S^1$ , let  $p: \mathbb{R} \rightarrow S^1$  s.t.  $p(t) = (\cos 2\pi t, \sin 2\pi t)$ .

Let  $(1,0) \in S^1$  be given. Then, for a neighborhood  $B_1(1,0) \cap S^1$ ,

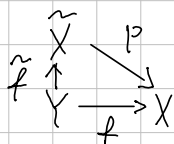
$$p^{-1}[B_1(1,0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} (n + (-\frac{1}{8}, \frac{1}{8}))$$

and for each  $n \in \mathbb{Z}$ ,

$$(n + (-\frac{1}{8}, \frac{1}{8})) \cong B_1(1,0) \cap S^1 \text{ via } p.$$



Def. Given a covering  $p: \tilde{X} \rightarrow X$  and a map  $f: Y \rightarrow X$ ,  
 a continuous map  $\tilde{f}: Y \rightarrow \tilde{X}$  such that  $p \circ \tilde{f} = f$  is called a lift of  $f$  to  $\tilde{X}$ ,  
 with respect to  $p$ .



Thm. Given a map  $F: Y \times I \rightarrow X$  and a map  $\tilde{F}: Y \times \{0\} \rightarrow \tilde{X}$  lifting  $F|_{Y \times \{0\}}$ , there is a unique map  $\tilde{F}: Y \times I \rightarrow \tilde{X}$  lifting  $F$  extended  $\tilde{F}: Y \times \{0\} \rightarrow \tilde{X}$ .  
Proof.

Def. Let  $p: E \rightarrow B$  be a covering map.

Given a continuous map  $f: X \rightarrow B$ , a lifting  $\tilde{f}: X \rightarrow E$  of  $f$  is a map satisfying  $p \circ \tilde{f} = f$ .

Thm. Let  $p: E \rightarrow B$  be a covering and  $p(e_0) = b_0$ .

For any path  $f: [0,1] \rightarrow B$  with  $f(0) = b_0$ ,

there exists a unique path  $\tilde{f}: [0,1] \rightarrow E$  satisfying  $p \circ \tilde{f} = f$

Proof, ~~불가능~~ 증명.

For any associated map by Homotopy of paths  $F: [0,1] \times [0,1] \rightarrow B$  with  $F(0,0) = b_0$ ,

there exists a unique associated map by Homotopy of paths  $\tilde{F}: [0,1] \times [0,1] \rightarrow E$

satisfying  $p \circ \tilde{F} = F$  and  $\tilde{F}(0,0) = e_0$ .